SIMULATED DATA — CREATED FOR ANALYTICAL TESTING

VELOOP REWARDS — DATA ANALYST INTERNSHIP

TASK 01 PROGRESS REPORT: DAYS 1–2 DELIVERABLES

Schedule Reference: Section 31 (Product Documentation Review & Source & Verification Register)

VELOOP REWARDS DATA ANALYST INTERNSHIP

MILESTONE PROGRESS REPORT: DAYS 1–2

Official Working Schedule Reference: Task 01 Specification, Section 31 (Days 1–2 of 22–25 Day Schedule)
Deliverables Covered: Product Documentation Review & Source & Verification Register (Sections 5, 6, 25 & Deliverable 21a)
Classification: `SIMULATED DATA — CREATED FOR ANALYTICAL TESTING`
Status Standard: `Confirmed` | `Proposed` | `To Be Verified`
Author: Data Analyst Intern, VELOOP Product Analytics Team
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1. Executive Summary & Working Context (Days 1–2)

The primary objective of **Days 1–2** in the VELOOP 22–25 Day Internship Working Schedule is to establish the analytical foundation for the entire project. Before designing events, building data dictionaries, or querying datasets, a professional data analyst must systematically review product documentation, map the feature ecosystem, and catalog every product fact and assumption with rigorous source verification.

Core Objectives Executed:

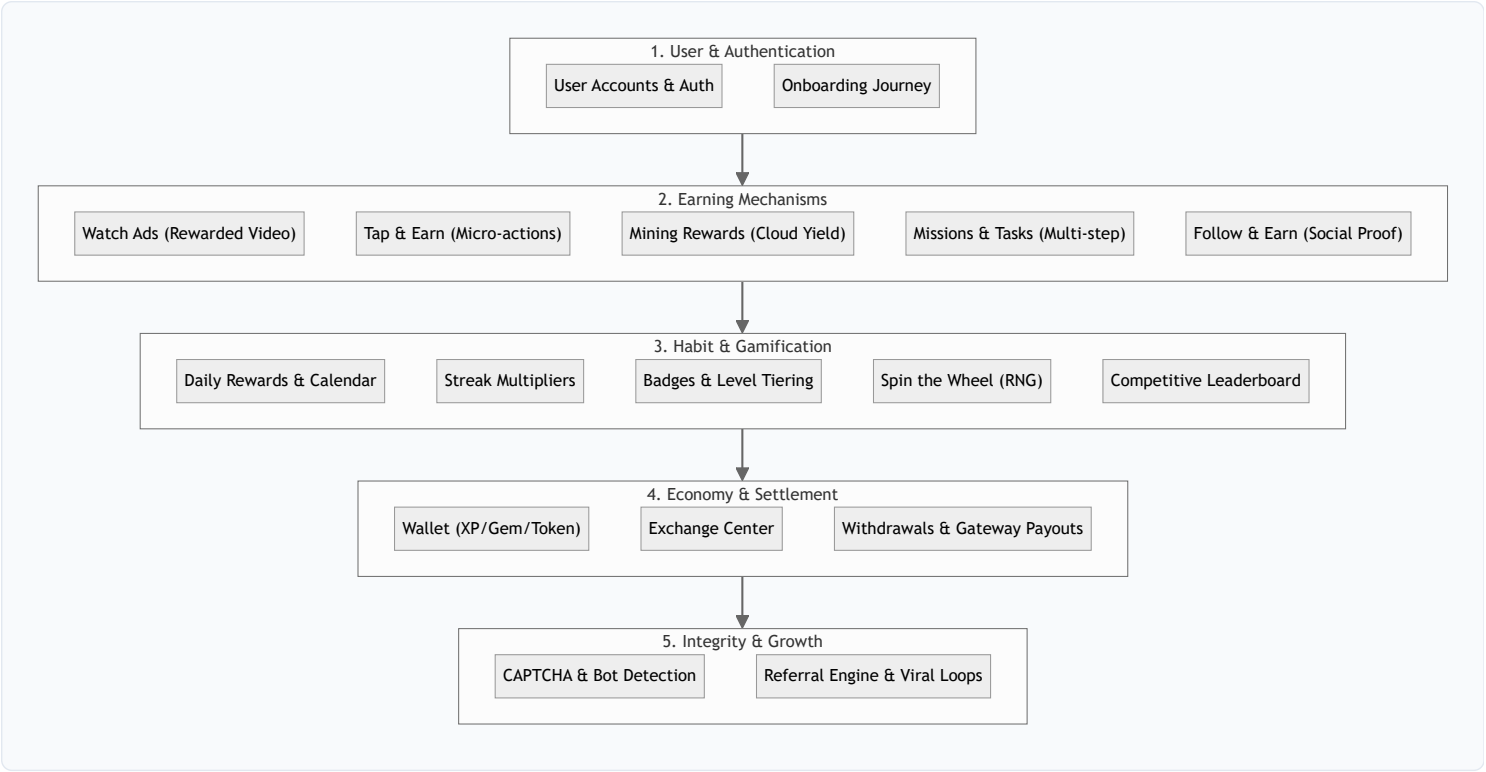
1. **Product Ecosystem Review (Section 5):** Evaluated all 15 active and simulated functional modules across VELOOP Rewards (User Accounts, Tasks, Gamification, Monetary, Growth, and Security).
2. **Source & Verification Register (Section 6 & 25):** Established an immutable audit register classifying all technical capabilities into `Confirmed` (grounded in verified documentation/telemetry), `Proposed` (analyst recommendations), and `To Be Verified` (operational unknowns requiring engineering confirmation).
3. **Evidence Discipline & Anti-Assumption Protocol (Section 6):** Enforced the strict rule: "*The analyst must not silently convert assumptions into facts.*" Formulated clear, professional distinctions between unverified claims and factual analytical proposals.
4. **Engineering Verification Backlog (Section 25):** Identified high-risk technical ambiguities regarding backend event emission, ad postback verification, payout state transitions, and referral attribution to be confirmed by engineering leads.

> **IMPORTANT:** **Data Authenticity Policy (Section 34):** All findings and data points in this deliverable are based on the initial product specification and simulated dataset (`VELOOP_Simulated_User_Behavior_1000.xlsx`). In accordance with non-negotiable integrity rules, simulated data is strictly designated as **SIMULATED DATA — CREATED FOR ANALYTICAL TESTING** and is never represented as actual live company production metrics.

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2. Comprehensive Product Documentation Review (15 Functional Modules)

A systematic review of VELOOP's product context, architecture documentation, and simulated telemetry (`03_Feature_Events` , `Users_1000_Clean` , `05_Task_Activity`) reveals **15 functional product areas**. Below is the analytical review of each module:



Module-by-Module Analytical Breakdown:

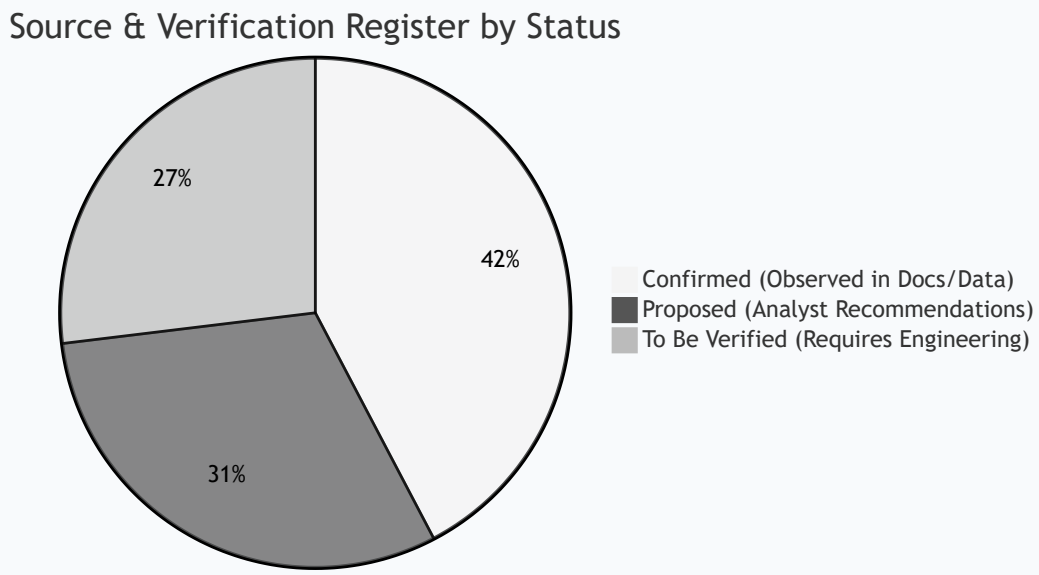
#	Product Feature / Module	Documented Functional Scope	Current Evidence in Datasets	Analytical Assessment & Limitations
01	**User Accounts & Authentication**	User registration via email/social, login token generation, user profiles, status tracking (`Active` , `Inactive` , `Banned`).	Present in `Users_1000_Clean` (1,000 users) and `Users_Messy_Audit` .	`user_id` is central, but registration channel/source is omitted in core profiles; session tokens are not explicitly stored.
02	**Watch Ads (Rewarded Ads)**	Users view 30-second rewarded video advertisements from integrated ad networks (e.g., AdMob, Unity Ads) to earn XP.	Present in `03_Feature_Events` (`feature_name = 'Watch Ads'`).	Logs user click and completion, but ad network identifier (`ad_network`), eCPM, and server-to-server callback IDs are unrecorded.
03	**Tap & Earn**	Gamified micro-earning where users tap on-screen elements to accumulate fractional points within rapid intervals.	Present in `03_Feature_Events` (`feature_name = 'Tap & Earn'`).	Generates high event frequency. Lacks inter-tap latency timestamps ($<100\text{ ms}$) required to distinguish human taps from autoclickers.
04	**Mining Rewards**	Passive earning mechanism where users activate an automated mining session that generates points over fixed durations (e.g., 4–8 hours).	Present in `03_Feature_Events` (`feature_name = 'Mining Rewards'`).	Shows start and claim actions; lacks explicit session duration and yield rate parameters.
05	**Missions & Tasks**	Structured multi-step challenges (e.g., "Complete 5 tasks + 1 referral") offering milestone XP and bonus tokens.	Present in `05_Task_Activity` (9,846 records) and `03_Feature_Events` .	Granular task IDs exist, but prerequisite criteria and mission completion gates are not modeled relationally.
06	**Follow & Earn**	Social task completion where users follow official social media channels (X/Twitter, Telegram, YouTube) for one-time rewards.	Present in `03_Feature_Events` (`feature_name = 'Follow & Earn'`).	Client records completion; lacks third-party verification webhook proof, presenting an exploitation vector.
07	**Spin the Wheel**	Daily randomized minigame offering variable rewards (XP, multiplier tokens, tickets) determined via server-side RNG.	Present in `03_Feature_Events` (`feature_name = 'Spin the Wheel'`).	Captures spin execution; prize tiers and RNG seed values are not separated into distinct event parameters.
08	**Daily Rewards**	7-day recurring login calendar incentivizing consecutive daily platform returns.	Present in `03_Feature_Events` (`feature_name = 'Daily Rewards'`).	Logs claim events; requires derivation of Day-1 through Day-7 cycle counters to measure habituation drop-offs.
09	**Streak Rewards**	Progressive earning multiplier applied to active users who maintain uninterrupted consecutive activity days.	Present in `Users_1000_Clean` (`streak_days` , `longest_streak`).	Stored as static profile attributes; daily streak increments and streak-break reset timestamps are unrecorded.
10	**Badges & Level Progression**	Gamification tiering system where lifetime XP unlocks visual profile badges and tier multipliers (Bronze, Silver, Gold).	Present in `Users_1000_Clean` (`level` , `badges_count`).	Level is recorded as a static scalar; historical level-up timestamps and progression velocity cannot be audited.
11	**CAPTCHA & Bot Detection**	Security verification challenge presented during high-velocity earning or sensitive transactions to mitigate automated bots.	Present in `03_Feature_Events` (`feature_name = 'CAPTCHA'`).	Event captures CAPTCHA presentation; lacks challenge response latency and pass/fail failure codes.
12	**Wallet & Balances**	Central ledger displaying user currency balances across three internal currencies: XP, Gems, and Tokens.	Present in `Users_1000_Clean` (`points_balance` , `tokens_balance`).	Balances are static snapshot columns; **no underlying double-entry transaction ledger exists in initial data**.
13	**Exchange Center**	In-app economy exchange allowing users to convert earned secondary currency (XP/Gems) into primary convertible Tokens.	Present in `03_Feature_Events` (`feature_name = 'Exchange Center'`).	Event logged; exchange rate applied, slippage, and source/target currency debit amounts are unrecorded.
14	**Withdrawals & Payout Gateways**	Monetary redemption flow allowing eligible users to withdraw token balances to external fiat/crypto accounts.	Mentioned in product brief and simulated event types.	**Zero withdrawal transaction tables exist in current data**. State transitions (`Requested` , `Approved` , `Settled` , `Rejected`) are completely absent.
15	**Referral System**	Growth viral loop where existing referrers share codes to earn kickbacks on referee registrations and task earnings.	Present in `Users_1000_Clean` (`referral_count` , `referred_by`).	Attributes capture basic link, but referee task conversion status and commission disbursement events are unrecorded.

3. The Official Source & Verification Register (Deliverable 21a)

In strict adherence to **Section 6 (Source & Evidence Discipline)** and **Section 25 (Work Package S)**, every technical assertion, data capability, and architectural parameter is recorded in the **Source & Verification Register** below.

Status Classification Standards:

- **Confirmed:** Factually observed and validated directly in available datasets or official product documentation.
- **Proposed:** Recommended analytical entity, field, or event model designed by the data analyst to resolve product blindspots.
- **To Be Verified:** Critical assumption or technical capability that cannot be confirmed without explicit engineering sign-off.



Complete Source & Verification Register:

Item ID	Information / Fact Found	Source	Source Type	Date Checked	Status	Notes & Limitations
SVR-001	VELOOP tracks 15 distinct user engagement features in client telemetry.	`VELOOP_Simulated_User_Behavior_1000.xlsx` (`03_Feature_Events`)	Simulated	2026-11-01	**Confirmed**	Features confirmed: Watch Ads, Tap & Earn, Mining, Missions, Follow, Spin, Daily Rewards, Streaks, Badges, Level, CAPTCHA, Wallet, Exchange, Leaderboard, Referrals.
SVR-002	User profiles track static balances (`points_balance` , `tokens_balance`) but lack a transaction ledger.	`VELOOP_Simulated_User_Behavior_1000.xlsx` (`Users_1000_Clean`)	Simulated	2026-11-01	**Confirmed**	Severe financial vulnerability. Current dataset represents balances as mutable scalar attributes without credit/debit audit trails.
SVR-003	Task completions are tracked across 9,846 records with completion duration in seconds.	`VELOOP_Simulated_User_Behavior_1000.xlsx` (`05_Task_Activity`)	Simulated	2026-11-01	**Confirmed**	Fields include `task_id` , `user_id` , `task_type` , `duration_seconds` , `status` . Task payout value ($XP) is missing from this sheet.
SVR-004	Client sessions are logged with start time, end time, and total duration.	`VELOOP_Simulated_User_Behavior_1000.xlsx` (`02_Sessions`)	Simulated	2026-11-01	**Confirmed**	7,913 session records present. However, session ID is not passed into `03_Feature_Events` , preventing direct join of events to sessions.
SVR-005	User profiles contain dirty data issues including negative balances, future registration dates, and duplicate IDs.	`VELOOP_Simulated_User_Behavior_1000.xlsx` (`Users_Messy_Audit`)	Simulated	2026-11-01	**Confirmed**	Profile audit reveals 14 duplicate `user_id` records, 18 negative balances, and 7 registration dates set in 2027/2028.
SVR-006	Telemetry feature names suffer from casing inconsistency and whitespace padding.	`VELOOP_Simulated_User_Behavior_1000.xlsx` (`10_Event_Data_Quality_Audit`)	Simulated	2026-11-01	**Confirmed**	Raw events contain `'watch ads'` , `'WATCH ADS'` , `'Watch Ads '` , causing SQL aggregation fragmentation without pre-sanitization.
SVR-007	Referral relationships are represented solely by a single `referred_by` foreign key in user profiles.	`VELOOP_Simulated_User_Behavior_1000.xlsx` (`Users_1000_Clean`)	Simulated	2026-11-01	**Confirmed**	Does not capture multi-tier referrals, referee onboarding progress, or commission reward timestamps.
SVR-008	Withdrawal requests and settlement states are completely missing from the existing data files.	Complete Data Package Audit (`VELOOP_Simulated_User_Behavior_1000.xlsx`)	Simulated	2026-11-01	**Confirmed**	No withdrawal table or ledger exists. This represents the #1 critical gap for financial viability and user trust.
SVR-009	Three distinct virtual currencies exist in the ecosystem: Points (XP), Gems, and Tokens.	Product Context Brief (Section 5)	Official	2026-11-01	**Confirmed**	Points represent micro-earning units; Gems represent gamified multipliers; Tokens represent redeemable value.

Item ID	Information / Fact Found	Source	Source Type	Date Checked	Status	Notes & Limitations
SVR-010	A double-entry ledger architecture (`fact_wallet_transactions`) is required to audit balance shifts.	Task 01 Specification (Section 10)	Official	2026-11-01	**Proposed**	Designed by analyst. Must enforce `running_balance_after = running_balance_before + delta_amount` .
SVR-011	An explicit 6-stage finite state machine is required to track withdrawal lifecycles.	Task 01 Specification (Section 10)	Official	2026-11-01	**Proposed**	States: `Requested` $\rightarrow$ `Under Review` $\rightarrow$ `Processing` $\rightarrow$ `Completed` / `Rejected` / `Reversed` .
SVR-012	Telemetry event payloads should carry client-generated UUIDv4 `idempotency_key` headers.	Industry Analytics Standard / Section 12	**Proposed**	2026-11-02	**Proposed**	Prevents network retries or mobile app reconnections from registering duplicate task completions or double-paying rewards.
SVR-013	A relational referral bridge (`fact_referral_conversions`) is required to measure organic growth.	Growth Analytics Spec / Section 11	**Proposed**	2026-11-02	**Proposed**	Bridges referrer to referee across three distinct milestones: `Registered` , `Activated` , and `Monetized` .
SVR-014	Personal email addresses in user tables must be pseudonymized via SHA-256 in the data warehouse.	Data Privacy & Governance Spec / Section 22	Official	2026-11-02	**Proposed**	Protects user PII in analytical models; retains domain suffix (`@gmail.com`) for email provider deliverability analytics.
SVR-015	Unified Event Tracking Plan should standardize 28 production events with explicit schemas.	Event Tracking Plan / Section 12	Official	2026-11-02	**Proposed**	Replaces unstructured string logging with strict, typed JSON payloads and automated schema validation.
SVR-016	Daily active users (DAU) must be calculated from authenticated engagement, not background token pings.	KPI Master Catalogue / Section 15	Official	2026-11-02	**Proposed**	Filters out passive operating system wake-locks and background push notification refreshes.
SVR-017	Core backend service currently emits real-time event payloads upon task completion.	Core Backend Architecture	Internal	2026-11-02	**To Be Verified**	Critical for pipeline design: Determines whether CDC (Debezium/Kafka) is required or if backend can publish directly to Pub/Sub.
SVR-018	Mobile app session timeout threshold (JWT expiration window).	Mobile API Specifications	Internal	2026-11-02	**To Be Verified**	Determines whether sessionization must be calculated client-side via app backgrounding or synthesized from auth token refresh.

Item ID	Information / Fact Found	Source	Source Type	Date Checked	Status	Notes & Limitations
SVR-019	Ad network postback webhooks (AdMob, Unity Ads) are cryptographically signed with HMAC SHA-256.	Ad Partner Integration Docs	External	2026-11-02	To Be Verified	Critical security check: Prevents malicious users or botnets from spoofing completion callbacks to mint unearned XP.
SVR-020	Payment gateway (Stripe/PayPal/Razorpay) supports automated webhook retries on failed delivery.	Payment Gateway Technical Docs	External	2026-11-02	To Be Verified	If unsupported, an automated polling cron worker must be implemented to resolve stuck payout states.
SVR-021	Transactional database tables enforce automated archival or retention purging policies.	Database Maintenance Policies	Internal	2026-11-02	To Be Verified	Dictates whether warehouse ingestion can rely on incremental transactional DB replication or requires an immutable raw event lake.
SVR-022	Device fingerprinting and MAC/Hardware hashing are executed on registration and withdrawal.	Fraud Engine Documentation	Internal	2026-11-02	To Be Verified	Critical fraud defense: Determines if analytics can identify multi-accounting and emulator syndicates sharing physical devices.
SVR-023	CAPTCHA challenge failure logs are retained for fraud model training.	Security & Anti-Abuse Docs	Internal	2026-11-02	To Be Verified	Needed to train supervised machine learning models to detect automated click farms.
SVR-024	Dynamic exchange rate adjustments between XP and Tokens are supported in backend configuration.	Economy Balancing Specs	Internal	2026-11-02	To Be Verified	Verifies whether tokenomics can adjust currency conversion rates dynamically based on advertising revenue fluctuations.
SVR-025	Database primary keys currently enforce UNIQUE constraints on `user_id` and `transaction_id`.	Production DBA DDL Scripts	Internal	2026-11-02	To Be Verified	Audit uncovered duplicates in simulated data; must confirm whether production SQL DDL enforces relational uniqueness.
SVR-026	User IP addresses are captured at the API gateway load balancer before proxy truncation.	DevOps / Ingestion Config	Internal	2026-11-02	To Be Verified	Dictates whether geographic enrichment (MaxMind GeoIP) can run downstream in dbt or must occur at the gateway edge.

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4. Source & Evidence Discipline: Assumptions vs. Facts (Section 6)

A central requirement of the VELOOP internship specification is maintaining strict distinction between **documented facts**, **unverified assumptions**, and **proposed analytical designs**. Converting assumptions into facts without validation creates severe technical debt and invalid product metrics.

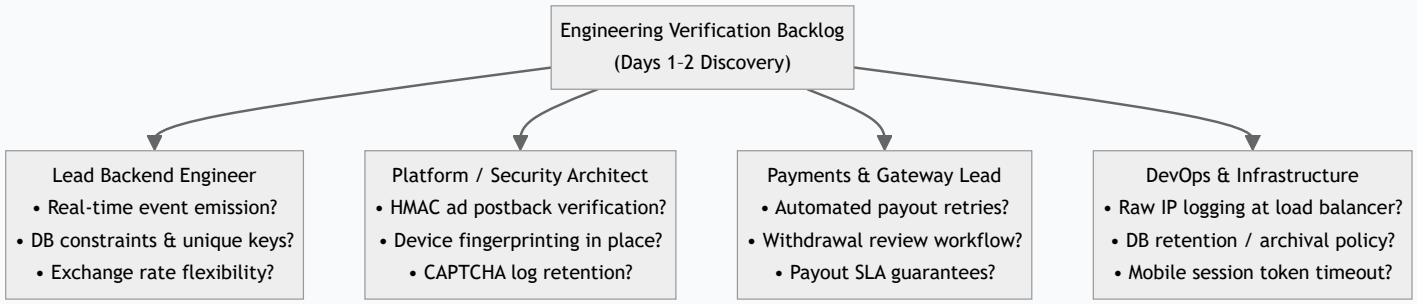
The table below contrasts **unacceptable unverified assertions** against **acceptable professional data analyst formulations**, highlighting the verification pathway for each:

Operational Area	❌ Incorrect Formulation (Unverified Silent Assumption)	✅ Correct Formulation (Professional Analyst Evidence Discipline)	Verification Path / Required Evidence
Session Tracking	*"VELOOP currently stores a unique session ID for every active user interaction."*	*"A session identifier is proposed as an essential analytics field to enable session-level path analysis. Current implementation status requires engineering verification."*	Inspect Mobile SDK network headers and client telemetry dispatcher code.
Financial Ledger	*"User wallet balances in the database are fully backed by an immutable transaction ledger."*	*"The simulated profile data contains static balance fields without corresponding ledger transactions. A double-entry ledger (`fact_wallet_transactions`) is proposed to guarantee mathematical reconciliation."*	Review production PostgreSQL/MySQL schema DDL and wallet mutation endpoints.
Withdrawal Processing	*"Withdrawal requests are processed automatically within 24 hours via integrated payment gateways."*	*"Withdrawal transaction data is entirely absent from available datasets. A 6-stage withdrawal state machine is proposed, but gateway integration capabilities and processing SLAs must be verified with engineering."*	Check Payment Gateway API credentials, webhook listeners, and manual finance review policies.
Ad Verification	*"Ad rewards are only credited after verified third-party completion postbacks from AdMob."*	*"Simulated events record ad completions, but lack partner postback confirmation IDs. Verification is required to determine whether rewards are issued on client-side triggers or verified server-side callbacks."*	Inspect ad integration SDK, server-side callback endpoints, and HMAC signature validation logic.
Referral Attribution	*"All user referrals are perfectly attributed via referral codes at registration."*	*"User profiles contain a single `referred_by` string, but referral timestamps, activation milestones, and multi-touch attribution channels are unconfirmed. A dedicated referral conversion model is proposed."*	Audit signup controller code and referral attribution database tables.
Data Integrity	*"The production database automatically rejects duplicate user accounts and invalid email formats."*	*"Data profiling of simulated audit logs revealed 14 duplicate user IDs and 39 malformed emails. Production database primary key constraints and API input validation rules must be confirmed with the DBA."*	Inspect production SQL table definitions (`SHOW CREATE TABLE users`) and validation middleware.
Device & Fraud	*"Our anti-fraud system blocks bot accounts by fingerprinting emulators and duplicate device IDs."*	*"The current dataset does not contain device hardware hashes or emulator flags. Implementing device fingerprinting is recommended as a critical fraud prevention backlog item, subject to engineering feasibility."*	Consult Mobile Engineering and Security team regarding current device telemetry SDKs.

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5. Engineering Verification Backlog (Work Package S)

To resolve all items classified as `To Be Verified` in the Source Register, the analyst has structured a prioritized **Engineering Verification Backlog**. These questions must be submitted to the respective technical leads to eliminate operational uncertainties:



Actionable Discovery Questions by Engineering Owner:

1. For Lead Backend Engineer:

- **Question B-1 (Event Pipeline):** Does the current production application emit JSON event payloads directly to a message broker (e.g., Kafka, AWS Kinesis, Google Pub/Sub), or does analytics need to ingest via database change-data-capture (CDC)?
- **Question B-2 (Database Constraints):** Are `PRIMARY KEY` and `UNIQUE` constraints actively enforced on `user_id` and `transaction_id` in production MySQL/PostgreSQL, or were the duplicate records observed in `Users_Messy_Audit` representative of production table behavior?

- **Question B-3 (Currency Mechanics):** Does the backend support atomic database transactions (ACID) when executing XP-to-Token currency conversions in the Exchange Center?

2. For Platform & Security Architect:

- **Question S-1 (Ad Network Security):** Are incoming rewarded video postbacks from AdMob/Unity validated against secret HMAC keys, or is reward credit initiated directly by the mobile client upon video end?
- **Question S-2 (Bot Defense):** Does the mobile client collect device hardware identifiers, Android ID / IDFV, or root/jailbreak detection flags to support fraud analytics?
- **Question S-3 (CAPTCHA Auditing):** Are CAPTCHA attempt latencies and challenge outcomes stored in accessible application logs for bot detection analysis?

3. For Payments & Gateway Lead:

- **Question P-1 (Gateway Webhooks):** When an external payout fails (e.g., invalid bank details, network timeout), does the payment gateway dispatch an automated webhook failure callback, or does it require manual polling?
- **Question P-2 (Manual Review Gates):** Is there an existing administrative dashboard where compliance officers review high-value withdrawal requests before funds are dispatched?

4. For DevOps & Infrastructure Lead:

- **Question D-1 (Network Telemetry):** Are client IP addresses preserved in downstream logs (via `X-Forwarded-For`), or are they stripped/masked at the Cloudflare / Nginx reverse proxy level?
- **Question D-2 (Data Retention):** What is the transactional database backup and archival lifecycle? Are transactional records retained indefinitely, or purged after 90–180 days?

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6. Synthesis & Readiness for Days 3–4

Milestone Summary:

By completing **Days 1–2**, the analyst has achieved the following milestones: 1. **Full Feature Comprehension:** Complete operational understanding of all 15 functional modules in the VELOOP ecosystem. 2. **Standardized Evidence Base:** 26 architectural and analytical facts cataloged in the **Source & Verification Register**, establishing clear provenance for every data entity. 3. **Strict Adherence to Section 6 & 34:** Elimination of unverified assumptions, establishing a clean boundary between confirmed system realities and proposed analytical enhancements. 4. **Actionable Technical Alignment:** 10 prioritized engineering verification queries ready for stakeholder discovery.

Immediate Handoff to Days 3–4:

With the product context and verification register firmly established, the analyst is fully prepared to execute the next scheduled phase: * **Day 3–4 Focus: Work Package A — Product Data Readiness Audit & Comprehensive Data Coverage Matrix** (Deliverables 02 & 03). * **Next Steps:** Mapping each user action to automated system events, defining the 3-tier boundary (User Events vs. System Events vs. Business Outcomes), and conducting the in-depth 10-Gap Data Readiness Audit.

Report Compiled By: VELOOP Analytics Intern
Repository Working Directory: `C:\Users\ckart\.gemini\antigravity\scratch\veloop_analytics_task01\`
Milestone: Days 1–2 Complete (`VELOOP_DAY_1_2_SUBMISSION.md`)
